

6. SLIDE SCANNER

Digital scanning of stained or unstained glass slides to generate high-resolution whole-slide images for review, documentation or analysis.

SOP No.: LAB-HISTO-06	Revision: 01	Effective: _____
Department: Histopathology / Research	Prepared by: _____	Approved by: _____

Purpose

- To convert glass-slide specimens into digital whole-slide images using a validated scanning workflow while preserving traceability and image quality.

PPE & Safety

- Handle slides carefully to avoid cuts and breakage. Keep scanner optics and slide-loading areas clean. Do not use liquids inside the scanner unless specifically permitted. Follow data-security and patient-confidentiality requirements.

Materials

- Clean, dry, labelled glass slides; compatible slide racks/trays; scanner workstation/software; validated storage location; calibration slide if required.

Procedure

- Power on the scanner and workstation. Check instrument status, available storage and software connectivity.
- Inspect slides for cleanliness, complete coverslips and correct labelling. Remove excess mounting medium and dust according to the lab procedure.
- Load slides in the correct orientation and select the validated scan profile/magnification.
- Verify case/specimen identifiers and image destination before starting.
- Start scanning. Monitor for loading errors, focus failures or incomplete scans. Do not open moving components unless the instrument indicates it is safe.
- Review scan quality: tissue coverage, focus, colour/staining appearance, artefacts and correct file association. Re-scan when required and document exceptions.

Quality Control

- Confirm image is complete, in focus and linked to the correct specimen. Periodically perform scanner calibration/quality checks according to manufacturer and laboratory schedules.

Cleaning & Shutdown

- Remove slides after completion. Clean accessible surfaces using the manufacturer's approved method. Do not touch optics with unapproved materials. Shut down or place in standby according to local policy.

Troubleshooting

- Focus failure → inspect slide/coverslip and focus settings. Colour mismatch → verify staining and scanner profile. Scan failure → check slide loading, storage and software status; escalate repeated errors to service support.

Important: This is a general SOP template. Before operational use, insert the exact manufacturer model, validated temperatures/times, reagent names, decontamination procedure, QC acceptance criteria and institutional biosafety/chemical-safety requirements.